Ownership Status, Diversification and Performance: Evidence from the Banking Industry in Sri Lanka

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Abstract

Decades of deregulations policies have helped Sri Lankan banks, both government owned and privately owned, to improve performance while diversifying their businesses. The purpose of this study is to investigate the relationship between diversification and performance of commercial banks, while taking into account the ownership status of these banks in in Sri Lanka. The emerging conclusion from this analysis is that, while private sector banks are significantly more diversified than their government-owned counterparts, both listed and unlisted, their performance is not necessarily superior to government-owned banks. This can be a result of the economic environment and the perception of the general public which has allowed the government-owned banks to entertain a significant market power over the private sector banks in the country.

Keywords

Diversification; Deregulation; State banks; Private banks; Bank performance; Ownership

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1. Introduction

Financial deregulation has encouraged commercial banks in many countries to diversify their business activities across a range of functions such as securities-related activities, insurance, foreign exchange, derivatives, investment management, financial planning, and many other off-balance sheet activities rather than relying on traditional loan making business. This contradicts with the popular argument that corporate diversification, in particular conglomerate diversification, reduces rather than enhances firm value. According to Jensen's (1986) free cash flow theory, managers of firms with high level of free cash flows and unused debt capacities tend to diversify their businesses beyond optimum levels with the objective of maximising their personal benefits at the expense of shareholders' wealth. The capital market, therefore, penalises diversification activities while rewarding focus increasing attempts (see for evidence, Eckbo, 1985; Sicherman and Pettway, 1987; Mørck, Shleifer, and Vishny, 990; Kaplan and Weisbach, 1992; Comment and Jarrrell, 1995; and John and Ofek, 1995). Empirically, Lang and Stulz (1994) determine that undiversified firms command an average Tobin's Q of 1.5 while diversified firms reporting a figure between 0.7 and 0.9. Berger and Ofek (1995) compare imputed values of the firm's components with the firm's total value and find that the value of the diversified firm is less than the sum of its parts by an average discount factor of 13 to 15 percent. Then, why do the commercial banks diversify their businesses? Is there a relationship between diversification and performance of commercial banks? A number of studies provide theoretical arguments both for and against these questions; the empirical evidence is also mixed. Moreover, the investigation of the relationship between bank diversification and performance is mainly limited to countries with developed financial systems such as the USA, Europe and Japan. The question of whether these relationships exist in recently liberalised financial systems remains largely under-investigated.

The purpose of our study is to investigate the relationship between diversification and performance of commercial banks, while taking into account the ownership status of these banks, using the data for a recently liberalised small economy, Sri Lanka which has successfully ended 30 years of internal conflict in 2009. As a result, economic activities are now booming. The banking sector appears to be fast growing with many new banks are being established and many existing banks are being listed on the Colombo Stock Exchange. As we explain in a later section, Sri Lanka was the first country in South Asia to implement a programme of economic and financial deregulation. These deregulation processes not only transformed the banking sector of the country, but also provided opportunities for commercial banks to engage in non-traditional business activities and thereby to have diversified businesses. Most of the banks with country-wide branch networks are owned either by the government or by the private sector. The operations of foreign banks are confined to the main business city of Colombo. [1] An interesting feature of private banks is that, except for recently established banks, all the banks owned by the private sector are listed on the Colombo stock exchange. A few government-owned banks are also listed on the exchange. This unique ownership structure can have an intervening influence on the relationship between diversification and performance as the motivation for diversification can vary across these categories of ownership. We are therefore motivated to address the following questions: (i) Is there a relationship between diversification and performance of Sri Lankan banks? (ii) Is this relationship moderated by the ownership of these banks?

The remainder of this paper proceeds as follows: Section II provides a brief review of literature. Section III describes the impact of financial deregulation on the growth of banking industry in Sri Lanka. Section IV discusses the sources of data while Section V outlines the methodology employed in the study. Section VI discusses the findings while the last section offers some conclusions.

2. Literature Survey and Hypotheses Development

Firms' diversification attempts are normally justified on the basis of the benefits associated with such attempts that include economies of scale (i.e. cost savings achieved from producing goods in large volume), economies of scope (i.e. benefits derived through the combination of marketing and distribution channels of different products), tax savings (i.e. offsetting losses of one division against the profits of another division), risk reduction (i.e. lowering the volatility of earnings/cash flows) and increase in debt capacity/leverage (i.e. increasing leverage and reducing cost of capital by lowering the probability of distress) (Berk and DeMarzo, 2007; Madura, 2012). Diversification also helps well established firms to exploit profitable new business opportunities (Gomas and Livdan, 2004). With respect to banking institutions, financial deregulation has been identified as the main driver of diversification. With the regulatory changes coming into effect, the competition in the banking industry has threatened the interest margins earned by banks forcing them to generate a substantial fraction of their income from non-loan activities rather than relying on traditional loan making business. Baele et al. (2007) discuss how the regulatory changes in the US and Europe encouraged banks to provide an array of financial services and how banks in those regions have responded to such changes with diversifying their businesses. It has been argued that diversified banks which engage in a variety of functions could benefit from the economies of scale and in consequence, improve their performance; such banks could achieve allocational efficiency through the internal capital markets available in financial conglomerates (Stein, 1997) and improve their profitability by sharing human capital, technologies and information (Sawada, 2013). Diamond (1984) finds that diversification reduces a bank's monitoring costs while Boyd and Prescott (1986) argue that a bank needs to be diversified as much as possible in order to optimise its value. However, an opposing argument is that functional diversification could affect bank performance negatively as it could aggravate the agency problem of a financial conglomerate (Jensen and Mecling, 1976; Rajan et al., 2000; Stein, 2002; Sawada, 2013). Laeven and Levine (2007) argue that the opaqueness of financial conglomerates make them difficult to monitor and therefore, a diversified bank could be prone to agency conflicts not only between insiders and outsiders, but also between the head office and divisional managers, among its divisions and between the bank and its customers.

Despite of many studies attempting to analyse the impact of diversification on bank performance, there appears to be no consensus among researchers (Martin and Sayrak, 2003). Acharya et al. (2002) find that diversification does not improve bank performance while Pilloff and Rhoades (2000) report that diversification attempts of banking institutions do not lead to any competitive advantages. While Laeven and Levine (2007) and Schmid and Walter (2009) uncover diversification discounts for financial conglomerates operating in Western Europe and the USA respectively, DeYong and Roland (2001) and Stiroh (2004) find revenue diversification to have a negative impact on the risk-return profile of US banks. However, studies such as Baele et al. (2007) and Chiorazzo et al. (2008) find that revenue diversification enhances the risk-return profiles of European banks. Further corroborative evidence that diversification improves bank performance can be found in Baele, De Jonghe and Vennet (2007), Elsas, Hackethal and Holzhäuser (2010) and Rossi, Schwaiger and Winkler (2009); while Baele et al. (2007) find income diversity to have a positive influence on the long-term value of the firm, Elsas et al. (2010) and Rossi, Schwaiger and Winkler (2009) find diversification to have a positive influence on bank profitability. Papanikolaou (2009) finds that the diversification of bank output enlarges efficiency margins associated with both cost and profit without changing the way banks treat risk. A reconciliatory finding for the debate of the influence of diversification on bank performance has been reported in Stiroh and Rumble (2006). The researchers find that even though the diversification is beneficial to US bank holding companies, these gains are offset by the increased exposure to non-interest activities which are much more volatile but not necessarily more profitable than interest-generating activities. These inconclusive findings motivate us to test the following hypothesis;

H_1: diversification will lead to improved performance of banks.

The argument of whether the bank performance influences diversification has not been subjected to extensive examination in empirical research. However, the research findings of Villalonga (2004) for non-financial sector reveal that the diversification does not necessarily destroy value. Venzin, Kumar and Kleine (2008), using a micro-level case study methodology, find that the diversification attempts of multinational European banks depend on their performance related attributes such as product portfolios, branch network configurations, branding strategies, social networks etc. Baele et al. (2007) argue that financial conglomerates can use the information accessed through lending relations to provide other financial services more efficiently implying that the bank performance is the cause of income diversification. Therefore, we test following hypothesis:

H_2: improved performance will lead banks to diversify their activities.

A number of studies have investigated the influence of bank ownership on their performance. It is a commonly accepted view among researchers that the government-owned banks perform poorly compared to private sector banks due to political interferences and their inbuilt inefficiencies (La Porta et al. 2002; Sapienza, 2004; Dinç, 2005; Micco et al., 2007). Claessens et al. (2002) argue that even if a bank is a publicly listed entity, such a bank could look for political benefits rather than efficiency improvements if the largest shareholder is the government. Lin and Zhang (2009) analyse Chine banks and find that the government-owned commercial banks are less profitable, less efficient, and have worse asset quality compared to private sector banks and foreign banks. Similarly, Micco et al. (2007) find that government-owned banks that operate in developing countries tend to have lower profitability and higher costs than privately owned banks; nevertheless, foreign banks operating in these countries appear to be more profitable than private banks. A study of Chinese banks by Berger et al. (2009) reveals that state owned big four banks are by far the least efficient and foreign banks are the most efficient. However, Bonin et al. (2005), who analysed 229 banks located in transition economies, conclude that privatisation is not sufficient to increase bank efficiency as government-owned banks are not always less efficient than domestic private banks; foreign ownership, however, leads to improved performance of banks in transition countries. With respect to the influence of ownership structure on diversification, Pennathur et al. (2012) find that public sector banks with higher levels of governmental ownership are significantly less likely to pursue non-interest income sources or diversification strategies. Based on this evidence we propose the following hypothesis:

> *H_3: ownership structure has an influence on both performance and diversification of banks.*

3. Deregulation and the Banking Sector Development in Sri Lanka

The decision taken by the government in late 1970s to liberalise the economy and to deregulate the financial services sector was a momentous policy initiative in the history of Sri Lanka. In this process, the Sri Lankan government introduced a number of policy changes to develop its financial system. These changes included the deregulation of interest rate determination, the introduction of market-based credit policies, relaxation of market entry for private sector and foreign banking institutions, and the introduction of necessary legislative measures to safeguard the financial system.This deregulation process resulted in an expansion of financial market activities allowing the banking sector of Sri Lanka to make notable gains not only in the country but also within the Asian Region.[2] The relaxation of market entry conditions brought about the much needed competition to the banking industry; the emergence of private sector financial institutions changed the landscape of the banking industry which had been dominated by government-owned banks prior to deregulation. Subsequent to deregulation, the dominant role of government-owned banks began to diminish as a result of the competition posed by newly established private banks. These private sector banks not only penetrated into the market dominated by the state banks, but also began to offer an array of fee-based services by making use of the opportunities provided by the liberalised economic environment. These diversification activities of private banks not only helped them expand their business activities across the country but also had a demonstrating impact on the state owned banks, encouraging them to diversify their activities into many areas in order to remain competitive in the industry.

The deregulation has also changed the ownership landscape of the banking industry in Sri Lanka. With the emergence of private sector banks after deregulation of the financial industry, five main categories of ownership can be found among the commercial banks operating in Sri Lanka: (i) state owned banks (for example, Bank of Ceylon), (ii) state owned listed banks (for example, Housing Development Finance Corporation of Sri Lanka)[3] , (iii) private sector listed banks (for example, Commercial Bank of Ceylon), (iv) private sector unlisted banks (for example, Amana Bank) and (v) foreign banks (for example, Deutsche Bank).

Table 1 provides some statistics relating to the development of the commercial banking sector in Sri Lanka. We can observe a substantial and continuous increase in the number of banks during the post deregulatory period. However, this increase can be attributable to the increase in domestic banks (both private sector and government-owned); even though the number of foreign banks has increased substantially following deregulation (from 7 to 17 in 1980), this number has remained unchanged over three decades after which a decline in the number of foreign banks can be observed in the last decade. The significance of state owned banks has decreased over the time; their contribution to the assets of the banking sector has declined from 92.43 percent in pre-deregulatory era to 38.97 percent in 2010 implying the increased role played by the private sector banks in the current banking industry in Sri Lanka. As the other statistics reported in Table 1 reveal, a significant improvement in banking activities can be observed with respect to the number of branches, bank deposits, bank assets and the credit provided by the banking sector.

4. Sample and Data

Currently, 37 banks conduct banking businesses in Sri Lanka having received the authority from the central bank of Sri Lanka to operate as licenced commercial banks. Out of these banks, we include banks for which the necessary data are available from the Bureau van Dijk BankScope database for at least five consecutive years. Such a data requirement was imposed considering the long-term nature of the diversification decision. Seventeen commercial banks fulfilled this requirement and therefore, we concentrate on those banks. This sample represents four government-owned banks, four government-owned listed banks and nine private sector listed banks.[4]

Table 1: Development of the Commercial Banking Industry in Sri Lanka

This table reports some statistics which are indicative of the development and the significance of the commercial banking industry in Sri Lanka. Sources: Annual reports of the Central Bank of Sri Lanka and the World Bank Database.

	1970	**1980**	**1990**	**2000**	**2010**
No. of commercial banks	10	23	28	35	37
- State owned banks	2	3	4	4	6
- State owned listed banks	0	0	0	2	3
- Private sector listed banks	0	0	0	4	8
- Private sector unlisted banks	1	3	6	8	8
- Foreign banks	7	17	18	17	12
State commercial bank assets to total commercial bank assets (%)	92.43	76.32	58.34	45.87	38.97
Commercial bank branches (per 100,000 adults)	9.65	10.54	12.59	14.10	16.73
Bank deposits to GDP (%)	14.12	22.64	19.22	31.15	31.06
Domestic credit provided by banking sector to GDP (%)	31.84	49.03	24.96	45.43	46.18
Deposit money bank assets to GDP (%)	15.23	19.67	21.26	32.15	33.88
Bank credit to bank deposits (%)	75.54	67.67	92.83	86.14	79.53

For this sample we collect the data on annual basis for the following variables for the period from 1997 to 2012 from BankScope database[5]: return on average assets, non-interest income, total income, total assets, book value of equity, total cost, reserves for impaired loans, total loans and the capital adequacy ratio. The data on the rate of inflation and the GDP growth rate were collected from the annual reports of the Central Bank of Sri Lanka.

5. Methodology

Influence of Bank Diversification on Performance:

Our measure of performance is the return on assets (ROA) and the measure of diversification is the ratio of non-interest income to total income (NIITI). In order to understand whether the bank performance is influenced by income diversification, we first estimate the following equation:

$$ROAA_{i,t} = \propto \beta_1(NITI_{i,t}) + \beta_2(LNTA_{i,t}) + \beta_3(ETA_{i,t}) + \beta_4(CI_{i,t}) + \beta_5(LLP_{i,t}) + \beta_6(GDPGR_{i,t}) + \beta_7(INFL_{i,t}) + \beta_8(War\ dummy_{i,t}) + \varepsilon_{i,t} \quad [1]$$

where, *ROAA* is the return on average assets, *NITI* is the non-interest income to total income, *LNTA* is the natural logarithm of total assets, *ETA* is the equity to assets, *CI* is the cost to income, *LLP* is the loan loss provision, *GDPGR* is the GDP growth rate, *INFL* is the rate of inflation and *War dummy* is the dummy representing separatist war period. The subscripts *i* and *t* represent bank *i* and year *t* respectively. The $\varepsilon_{i,t}$ is the error term. The control variables were included in the regression model because they have been used in prior studies (see, for example, Baele et al., 2007; and Berger et al., 2010) due to the following reasons:

- the natural logarithm of 'total assets' to capture size-related influences that stem from differences in size among banks, and to control for firm size effects (for example Fama and French, 1992).

- Equity to assets (the book value of equity divided by total assets) to capture the influence of bank capital on firm performance.
- Cost to income (total cost divided by total income) to capture the influence of operational efficiency on bank performance.
- Loan loss provisions (the reserves for impaired loans divided by total loans) to capture the quality of loans in the portfolio of the bank's assets.
- GDP growth rate and the rate of inflation to capture the macro-economic influence on the performance of banks.
- War dummy to capture the impact of separatist war prevailed in some years of the study period.[6]

We then consider whether the influence of NITI on ROAA is conditional to the ownership status of a bank. For this purpose, we add a dummy variable representing government-owned banks, both listed and unlisted, to equation [1] and estimate the following model:

$$ROA_{i,t} = \propto + \beta_1(NITI_{i,t}) + \beta_2(LNTA_{i,t}) + \beta_3(ETA_{i,t}) + \beta_4(CI_{i,t}) + \beta_5(LLP_{i,t}) + \beta_6(GDPGR_{i,t}) + \beta_7(INFL_{i,t}) + \beta_8(War\ dummy_{i,t}) + \beta_9(GOV\ dummy_{i,t}) + \varepsilon_{i,t} \qquad [2]$$

where GOV $\text{dummy}_{i,t}$ takes the value of one if the bank is a government-owned entity and zero otherwise.

We further split government-owned banks into two groups as government-owned unlisted banks (GOVU) and government-owned listed banks (GOVL) and estimate the following equation to investigate if being government-owned and listed on the stock exchange makes a difference to the above relationship:

$$ROA_{i,t} = \propto + \beta_1(NITI_{i,t}) + \beta_2(LNTA_{i,t}) + \beta_3(ETA_{i,t}) + \beta_4(CI_{i,t}) + \beta_5(LLP_{i,t}) + \beta_6(GDPGR_{i,t}) + \beta_7(INFL_{i,t}) + \beta_8(War\ dummy_{i,t}) + \beta_9(GOVU\ dummy_{i,t}) + \beta_9(GOVL\ dummy_{i,t}) + \varepsilon_{i,t} \qquad [3]$$

where $GOVU_{i,t}$ takes the value of one if the bank is government-owned but unlisted and zero otherwise, and $GOVL_{i,t}$ takes the value of one if the bank is government-owned and listed on the exchange and zero otherwise.

Influence of Bank Performance on Diversification:

We first estimate the following equation to investigate whether the income diversification of a bank is influenced by its performance:

$$NITI_{i,t} = \propto + \beta_1(ROA_{i,t}) + \beta_2(LNTA_{i,t}) + \beta_3(LTA_{i,t}) + \beta_4(LLP_{i,t}) + \beta_5(CAPAD_{i,t}) + \beta_6(GDPGR_{i,t}) + \beta_7(INFL_{i,t}) + \beta_8(War\ dummy_{i,t}) + \varepsilon_{i,t} \qquad [4]$$

where, $LTA_{i,t}$ is the total loans to total assets ratio and $CAPAD_{i,t}$ is the capital adequacy ratio. The other variables are defines as before. The control variables are included in the model for the following purposes:

- Natural logarithm of total assets (*LNTA*) to capture the influence of firm size on diversification.
- Loans to assets ratio (*LTA*) to capture the influence of the concentration on loan making on diversification.
- Loan loss provisions (*LLP*) to capture the influence of loan quality on diversification.
- Capital adequacy ratio (*CAPAD*) to capture the influence of risk and solvency on the bank on diversification.

We then examine whether the influence of performance on the diversification is conditional to the ownership status of the bank. For this purpose, as we did in performance regressions, the following two equations are estimated:

$$NITI_{i,t} = \propto + \beta_1(ROA_{i,t}) + \beta_2(LNTA_{i,t}) + \beta_3(LTA_{i,t}) + \beta_4(LLP_{i,t}) + \beta_5(CAPAD_{i,t}) + \beta_6(GDPGR_{i,t}) + \beta_7(INFL_{i,t}) + \beta_8(War\ dummy_{i,t}) + \beta_9(GOV\ dummy_{i,t}) + \varepsilon_{i,t} \quad [5]$$

$$NITI_{i,t} = \propto + \beta_1(ROA_{i,t}) + \beta_2(LNTA_{i,t}) + \beta_3(LTA_{i,t}) + \beta_4(LLP_{i,t}) + \beta_5(CAPAD_{i,t}) + \beta_6(GDPGR_{i,t}) + \beta_7(INFL_{i,t}) + \beta_8(War\ dummy_{i,t}) + \beta_9(GOVU\ dummy_{i,t}) + \beta_9(GOVL\ dummy_{i,t}) + \varepsilon_{i,t} \quad [6]$$

6. Findings

i. Sample Characteristics, Diversification and Performance by Bank Ownership:

Table 2 reports descriptive statistics for sample characteristics by bank ownership. An average Sri Lankan bank employed assets worth of 588187 million rupees (the average of natural logarithm of total assets = 13.28) during this 16-year period of which 61 percent accounted for loans given to customers. Government-owned unlisted banks are the largest banks, followed by private sector listed banks and government-owned listed banks. Government-owned listed banks reported the highest loans to assets ratio of 66.30 percent which is higher than the overall average of 61 percent implying that they tend to be more generous in providing loans to customers than their government-owned unlisted and private sector listed counterparts. Similarly, government-owned listed banks seem to maintain a stronger equity base, reporting the highest equity to assets ratio of 16.62 percent, compared to other two categories. The private sector listed banks are the least cost efficient, reporting the highest cost to income ratio of 67.27 percent; but, they seem to be more efficient in collecting outstanding loans than the other two categories as evidenced by the lowest loan loss provision ratio of 3.60 percent. The government-owned banks, both listed and unlisted, maintain a higher capital adequacy ratio than their private sector counterparts.

Figures 1 and 2 display the behaviour of return on assets (ROA) and non-interest income to total income (NITI) respectively for the three types of banks – government-owned banks (GOV), government-owned listed banks (GOVL) and private sector listed banks (PRIVL) – for the 16-year period. As Figure 1 reveals, the profitability of Sri Lanka banks has remained highly volatile during this period probably due to the negative impact of separatist war on the performance of banks in some years of the study period. However, government-owned listed banks show a superior performance in most of the years of the sample period followed by private sector listed banks; government-owned unlisted banks record the least performance. In Figure 2, the non-interest income share of government-owned unlisted banks and government-owned listed banks remain more stable compared to that of the private sector listed banks.

Table 2: Sample characteristics by bank ownership: 1997-2012

This table reports descriptive statistics for a number of variables for the full sample as well as for the sub samples classified on the basis of their ownership. The three sub samples included are (i) government owned unlisted banks, (ii) government owned listed banks and (iii) private sector listed banks. The definitions of variables are as follows: LNTA: Natural Logarithm of Total Assets; ETA: Equity to Total Assets; LTA: Total Loans to Total Assets; CI: Cost to Total Income; LLP: Loan Loss Provisions to Total Loans; and CAPAD: Capital Adequacy Ratio.

Type of bank	Statistic	LNTA	ETA	LTA	CI	LLP	CAPAD
All banks	Mean	13.2846	0.1078	0.6093	0.6146	0.0405	0.1483
	Median	13.4229	0.0829	0.6291	0.6129	0.0330	0.1387
	Std. deviation	1.3416	0.0702	0.1230	0.1619	0.0281	0.0472
	Minimum	10.8837	0.0212	0.2950	0.3253	0.0063	0.0823
	Maximum	15.2231	0.2602	0.8345	0.9554	0.1162	0.2615
Government owned unlisted banks	Mean	14.2657	0.0854	0.5576	0.5648	0.0493	0.1542
	Median	14.7043	0.0527	0.5479	0.5879	0.0368	0.1372
	Std. deviation	1.1003	0.0726	0.1618	0.1718	0.0396	0.0580
	Minimum	11.3826	0.0212	0.2950	0.3253	0.0063	0.0823
	Maximum	15.2231	0.2602	0.8345	0.9554	0.1162	0.2615

Government owned listed banks	Mean	12.4154	0.1662	0.6630	0.5008	0.0407	0.1617
	Median	12.8527	0.1589	0.6585	0.4708	0.0371	0.1555
	Std. deviation	1.0634	0.0567	0.1150	0.1680	0.0219	0.0474
	Minimum	10.8837	0.0212	0.4013	0.3253	0.0087	0.0823
	Maximum	14.1303	0.2602	0.8345	0.9554	0.0996	0.2615
Private sector listed banks	Mean	13.0538	0.1017	0.6192	0.6727	0.0360	0.1415
	Median	13.3044	0.0791	0.6328	0.6525	0.0315	0.1335
	Std. deviation	1.2465	0.0636	0.0905	0.1263	0.0212	0.0397
	Minimum	10.8837	0.0212	0.2950	0.4059	0.0063	0.0823
	Maximum	15.2095	0.2602	0.8345	0.9554	0.1162	0.2615

Table 3: Comparison of Diversification and performance by bank ownership: 1997-2012

This table reports descriptive statistics for the diversification variable (NIITI) and the performance variable (ROA) for the full sample as well as for the sub samples classified on the basis of their ownership. The three sub samples included are (i) government owned banks, (ii) government owned listed banks and (iii) private sector listed banks. The definitions of variables are as follows: NIITI: Non-interest Income to Total Income; and ROA: Return on Assets. The last column reports *F*-statistic, Kruskal-Wallis statistic and Levene statistic that test the differences in means, medians and standard deviations respectively across three sub samples. *, **, and *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Variable	**Statistic**	**All banks**	**Government owned unlisted banks**	**Government owned listed banks**	**Private sector listed banks**	**Test statistic**
Non-interest Income to Total Income (NIITI)	Mean	0.3274	0.2758	0.3304	0.3522	*F* statistic = 5.67***
	Median	0.3304	0.2876	0.3330	0.3377	*Kruskal-Wallis* statistic = 10.18***
	Std. deviation	0.1283	0.1454	0.0547	0.2423	*Levene* statistic = 5.01***
Return on Assets (ROA)	Mean	0.0151	0.0128	0.0207	0.0145	*F* statistic = 3.76**
	Median	0.0127	0.0101	0.0214	0.0120	*Kruskal-Wallis* statistic = 12.14***
	Std. deviation	0.0140	0.0121	0.0148	0.0143	*Levene* statistic = 0.75

However, private sector listed banks seem to generate a higher proportion of their income from non-interest sources compared to government-owned listed and unlisted banks.

Table 3 compares diversification and performance among banks on the basis of their ownership status. As we observed in Figure 2, private sector listed banks generate the highest proportion of income from non-interest sources (mean = 35%; median = 34%). The non-interest income proportion of government-owned listed banks (a mean/median ratio of 33%) is slightly lower than that reported by private sector banks; this ratio for the government-owned unlisted banks remain at a much lower level (mean = 28%; median = 29%). Clearly, banks listed on the stock exchange and owned either by the government or the private sector investors seem to take the advantage of opportunities provided by the deregulated financial system and offer diversified products to their customers than those totally owned by the government. This may be a result of listed banks being monitored by the investment community which forces them to make use of opportunities available in the deregulated economy. We observe a similar picture with respect to performance of these banks. The best performing banks are the government-owned listed banks; they report an average ROA of 2.07 percent followed by the private sector listed banks (1.45%) and government-owned unlisted banks (1.28%). The statistics reported in the last column of Table 2 reveal that statistically significant differences existed among the three groups of banks with respect to both diversification and performance. Such a finding provides a strong motivation to investigate the relationship between diversification and performance subject to the ownership status of Sri Lankan banks. This is addressed in the forthcoming sections.

ii. The effect of bank diversification on performance:

Table 4 reports the estimates of regression equations [1] to [3]. Clearly, Sri Lankan banks have benefitted from income diversification; the coefficient on NITI is positive (ranging from 0.0086 to 0.0095) and significant at the 5 percent level in all three models estimated. However, the 'government-owned bank (GOV) dummy' in column 2 enters into equation [2] with a negative coefficient (-0.0055) which is significant at the 5 percent level. This implies that, compared to their private sector counterparts, government-owned banks underperform in terms of profitability by a significant margin. When these government-owned banks were disaggregated into two groups as 'government-owned unlisted (GOVU) banks' and 'government-owned listed (GOVL) banks', the results reported in column 3 reveal that 'GOVU dummy' enters into equation [3] with a negative and significant coefficient (coefficient = -0.0058; *t*-value = -2.55) while 'GOVL dummy' enters with an insignificant coefficient (coefficient = -0.0051; *t*-value = -1.04). This finding clearly indicates that the profitability performance of government-owned listed banks is not dissimilar to that of private sector listed banks but government-owned unlisted banks demonstrate a significant underperformance.

Table 4: The effect of bank diversification on performance

This table reports the regression outputs for equations [1] to [3]. *, **, and *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Independent Variable	**Coefficient (t-stat.)**	**Coefficient (t-stat.)**	**Coefficient (t-stat.)**
Constant	0.0145	0.0167	**0.0176**
	(1.29)	(1.42)	**(1.23)**
Non-interest income to total income (NIITI)	0.0095**	0.0093**	0.0086**
	(2.53)	(2.03)	(2.00)
Natural logarithm of total assets (LNTA)	0.0007	0.0008	0.0008
	(0.97)	(1.11)	(0.83)
Equity to assets (ETA)	0.0862***	0.0893***	0.0861***
	(7.56)	(7.67)	(7.22)
Cost to income (CI)	-0.0310***	-0.0328***	-0.0328***
	(-7.15)	(-5.45)	(-5.11)
Loan loss provision (LLP)	-0.0365*	-0.0311	-0.0332*
	(-1.92)	(-1.54)	(-1.63)
GDP growth rate (GDPGR)	0.0445	0.0413	0.0412
	(1.21)	(1.11)	(1.06)
Rate of inflation (INFL)	-0.0149*	-0.0139*	-0.0143*
	(-1.78)	(-1.80)	(-1.84)
War period dummy (WAR dummy)	-0.0026**	-0.0027**	-0.0027**
	(-2.57)	(-2.28)	(-2.22)

Government owned bank dummy (GOV dummy)		-0.0055** (-2.26)	
Government owned unlisted bank dummy (GOVU dummy)			-0.0058** (-2.55)
Government owned listed bank dummy (GOVL dummy)			-0.0051 (-1.04)
R-squared	0.53	0.55	0.54
Model F-statistic	28.59***	27.55***	23.75***
Number of observations	213	213	213

Turning to control variables, we find that the coefficients of equity to assets ratio are positive and significant in all three models estimated implying that well capitalised banks are generally associated with better performance. The size of the bank does not have any influence on bank performance, however. On the other hand, costs to income and loan loss provisions have negative influences on bank ROA; banks that are inefficient in managing their costs and those that are generous in providing loans (and therefore have to allocate a high portion of their income to loan loss provisions) are less effective in achieving higher profitability. Among macroeconomic factors, we find that the rate of inflation has a significant negative influence on bank profitability; high inflation is normally associated with high interest rates in the economy and this discourages borrowers imparting a negative influence on bank performance. Not surprisingly, we find that war period dummy enters into regression models with negative and significant coefficients; the contraction of economic activities during the war period has had a negative impact on bank profitability. The model R^2 values are high ranging between 53 percent and 55 percent and the model F-statistics are significant at the 1 percent level.

iii. *The effect of bank Performance on diversification*:

In this section, we investigate whether the income diversification is influenced by bank performance and, if so, the influence of performance on diversification is conditional on the ownership status of a bank. For this purpose, regression equations [4] to [6] are estimated and the resultant outputs are reported in Table 5. The coefficients of 'return on assets (ROA)' variable are positive (ranging from 2.2253 to 2.4160) and significant at the 1 percent level implying that highly profitable banks have a tendency to achieve a high level of diversification in their income. However, as reported in column 2, the GOV dummy enters into equation [5] with a negatively significant coefficient (coefficient = -0.0541; t-value = -2.03) implying that government-owned banks are significantly less diversified than their private sector counterparts. When the government-owned banks are split into two groups as unlisted and listed, we find that this negative influence stems from 'government-owned unlisted banks' as implied by the significant negative coefficient generated for the GOVU dummy; the income diversification of 'government-owned listed banks' is not significantly different from that of private sector listed banks.

Turning to control variables, we find that both bank size and capital adequacy ratio generate negative and significant coefficients implying that large banks and those that maintain a high capital adequacy ratio tend to less diversify their income. The macro-economic variables as well as the war period do not have any significant impact on income diversification attempts of these banks.

iv. *Performance-diversification relationship after controlling for endogeneity*:

The analyses conducted so far point to a possible causality between bank performance (ROA) and income diversification (NITI); in equations [1] to [3], NIITI variable generates positive and significant coefficients while in equations [4] to [6], ROA variable generates positive and significant coefficients. In order to take into account this possible causal effect, in this section, we estimate equations [3] and [6] simultaneously by employing 2SLS methodology. Our objectives here are (i) to examine whether there is a causal relationship between diversification and performance and (ii) whether the influence of bank ownership status on the performance-diversification relationship holds even after controlling for this possible the two-way relationship.

The output of 2SLS regression is reported in Table 6. Both NITI and ROA coefficients remain positive and significant implying a causal relationship between these two variables; greater diversification is associated with higher profitability and higher performance is associated with greater diversification. The influence of control variables on performance remain similar to what we observed before. The only exceptions are the significant positive coefficients generated for bank size and GDP growth in ROA regression implying that bank size and economic expansion have positive influences on bank performance. In our NITI regression of 2SLS system, all the control variables except 'loan loss provision' remain insignificant implying that both bank-specific and macro-economic variables do not have any significant impact on diversification activities of these banks; diversification is affected solely by the performance of the bank. The influence of ownership status of banks on both performance and diversification alters when the equations are estimated simultaneously. Relative to their private sector listed counterparts, government-owned banks, both listed and unlisted, are significantly less diversified; both GOVU dummy and GOVL dummy generate negative and significant coefficients (-0.0610 and -0.0446) coefficients in NITI regression. However, the performance of private sector listed banks is not significantly different from those of government-owned listed and unlisted banks; in ROA regression, both GOVU dummy and GOVL dummy generate insignificant coefficients. The emerging conclusion from this analysis is that, even though private sector listed banks grab the opportunities provided by the liberalised economy by diversifying their business activities into non-interest generating sources, such a transformation has not necessarily helped them improve their profitability performance by a significant margin.

v. *Discussion*:

Taken together, our analyses provide three main findings: (i) there is a bidirectional relationship between diversification and performance of Sri Lankan banks, (ii) privates sector banks are significantly more diversified than government-owned (both listed and unlisted) banks and (iii) the performance of government-owned banks (both listed and unlisted) is not dissimilar from that of private sector banks. All these relationships are plausible due to a number of reasons. First, the two-way relationship between diversification and performance is justifiable since the analyses are conducted for the most recent years of the post-deregulatory period in which banks were offered with opportunities to diversify their sources of income and improve performance. Second, the government-owned banks are less diversified than private sector banks since they tend to pursue political objectives. Many senior officers of government-owned banks are appointed by the government and therefore the government has a significant influence on policy decisions of these banks. For example, many government-sponsored credit and investment programmes are implemented through these banks and these banks are expected to give a high priority to these government-sponsored programmes inhibiting their motivation to generate a substantial income from non-loan sources. As Claessens et al. (2002) argue, even if a bank is a listed entity, if the largest shareholder is the government, the government could look for political benefits rather than efficiency improvements. Finally, the performance of private sector banks is not dissimilar to government-owned banks because of the market power entertained by government-owned banks. Irrespective of financial deregulation, the Sri Lankan economy has been dominated by the government sector and the common perception among the population is that the government sector is superior to the private sector. This has helped government-owned banks to maintain a substantial market power in the banking sector of the country and to benefit from such an advantageous position. Additionally, Sri Lanka is still a developing country and the financial literacy among the general public is still low. As a result, new products and services offered by the private sector banks are not penetrating into the general public and many are still comfortable with those traditional products offered by the government sector banks.

7. Conclusion

In this study, we analyse the relationship between diversification and performance of Sri Lankan banks using the data for three types of domestic banks (i.e. government-owned unlisted banks, government-owned listed banks and private sector listed banks) for the period from 1997 to 2012. Our initial analyses revealed the following: First, the performance of Sri Lankan banks has been significantly improved by their diversification attempts. Second, the performance of government-owned unlisted banks is significantly lower than that of the private sector listed banks while the performance of government-owned listed banks is not dissimilar to that of the private sector listed banks. Third, the performance of Sri Lankan banks has a significant positive influence on their diversification of sources of income. Finally, government-owned unlisted banks are significantly less diversified than private sector listed banks while the income diversification of government-owned listed banks is not significantly different from that of private sector listed banks.

When the two-way relationship between diversification and performance was scrutinised employing 2SLS technique, it was found that a statistically significant bidirectional relationship exists between diversification and performance. We attribute this to the opportunities provided by the post-deregulatory era of the country for the banking sector. However, when this two-way relationship was taken into account, it was found that while private sector banks are significantly more diversified than their government-owned counterparts, both listed and unlisted, their performance is not necessarily superior to government-owned banks. This can be a result of the economic environment and the perception of the general public which has allowed the government-owned banks to entertain a significant market power over the private sector banks in the country.

Table 5: The effect of bank Performance on diversification

This table reports the regression outputs for equations [4] to [6]. *, **, and *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Independent Variable	Coefficient (t-stat.)	Coefficient (t-stat.)	Coefficient (t-stat.)
Constant	0.7920***	0.7492***	0.7612***
	(5.33)	(5.18)	(4.89)
Return on assets (ROA)	2.3274***	2.4160***	2.2253***
	(2.95)	(3.16)	(2.78)
Natural logarithm of total assets (LNTA)	-0.0234**	-0.0207**	-0.0201**
	(-2.48)	(-2.27)	(-1.96)
Loans to assets (LTA)	-0.1307	-0.1087	-0.1288
	(-1.60)	(-1.36)	(-1.57)
Loan loss provisions (LLP)	-0.1698	-0.0454	-0.1205
	(-0.51)	(-0.14)	(-0.36)
Capital adequacy ratio (CAPAD)	-0.6200***	-0.5405***	-0.5514***
	(-3.00)	(-2.64)	(-2.62)
GDP growth rate (GDPGR)	-0.5507	-0.5740	-0.5674
	(-0.87)	(-0.91)	(-0.89)
Rate of inflation (INFL)	0.1905	0.1980	0.1990
	(1.29)	(1.34)	(1.34)
War period dummy (WAR dummy)	0.0030	0.0039	0.0025
	(0.17)	(0.21)	(0.14)

Government owned bank dummy		-0.0541**	
(GOV dummy)		(-2.03)	
Government owned unlisted bank			-0.0611*
dummy (GOVU dummy)			(-1.77)
Government owned listed bank			-0.0443
dummy (GOVL dummy)			(-1.12)
R-squared	0.09	0.11	0.10
Model F-statistic	2.50**	2.72***	2.34**
Number of observations	213	213	213

Table 6: 2SLS Regression Estimates

This table reports the 2SLS output when equations [3] and [6] are estimated simultaneously. *, **, and *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Independent Variable	Coefficient (t-stat.)	Coefficient (t-stat.)
Constant	-0.0262**	0.4313***
	(-2.18)	(3.25)
Non-interest income to total income (NIITI)	0.0960**	
	(2.53)	
Return on assets (ROA)		2.9121***
		(3.51)
Natural logarithm of total assets (LNTA)	0.0014**	-0.0092
	(2.26)	(-1.17)
Equity to assets (ETA)	0.0846***	
	(3.21)	
Cost to income (CI)	-0.0324***	
	(-5.75)	
Loan loss provisions (LLP)	-0.0320	0.5831*
	(-1.12)	(1.75)
Loans to assets (LTA)		0.0510
		(0.66)
Capital adequacy ratio (CAPAD)		-0.2169
		(-1.05)

GDP growth rate (GDPGR)	0.1077**	-0.8061
	(2.01)	(-1.09)
Rate of inflation (INFL)	-0.0250*	0.1989
	(-1.78)	(1.11)
War period dummy (WAR dummy)	-0.0029**	0.0106
	(-2.23)	(0.49)
Government owned unlisted bank dummy	0.0020	-0.0610***
(GOVU dummy)	(0.65)	(-2.73)
Government owned listed bank dummy	-0.0020	-0.0446*
(GOVL dummy)	(-0.80)	(-1.83)
R-squared	0.73	0.15
Model F-statistic	55.02***	3.53***
Number of observations	213	213

Figure 1: Return on Assets (ROA): 1997-2012

This figure displays average annual Return on Assets (ROA) for three sub samples from 1997 to 2012. The three sub samples included are (i) government owned banks (GOV), (ii) government owned listed banks (GOVL) and (iii) private sector listed banks (PRIVL).

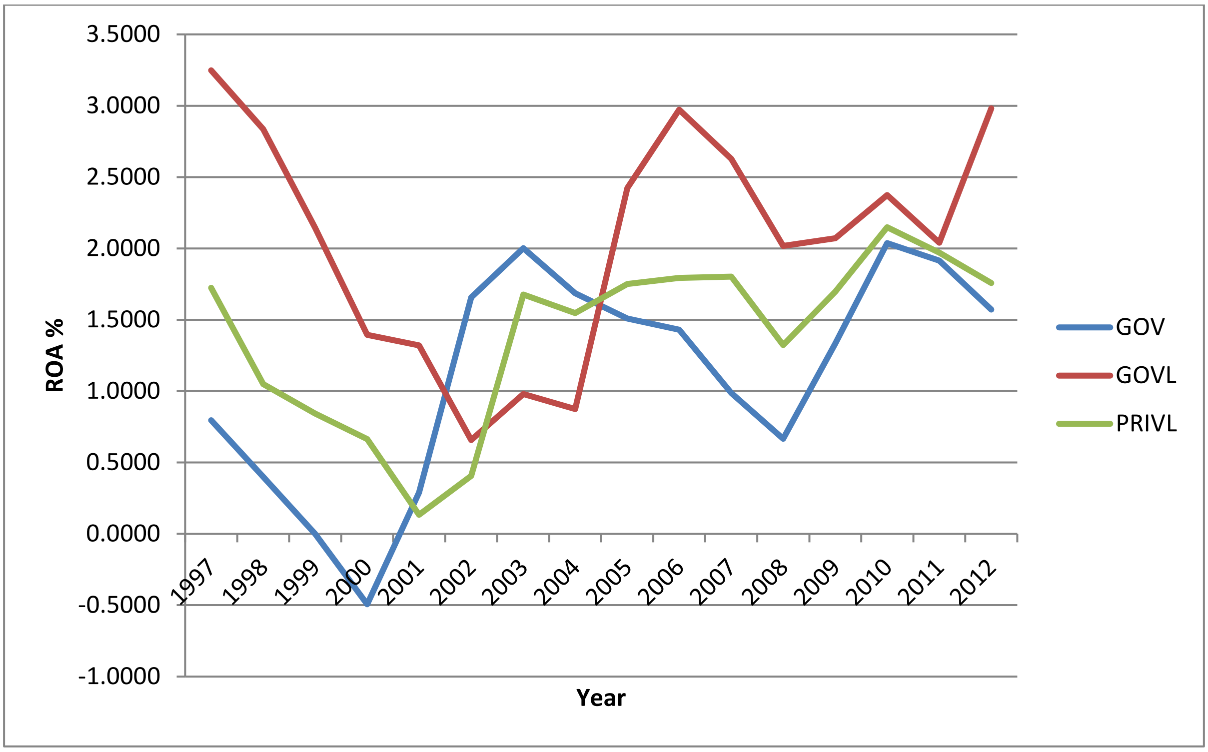
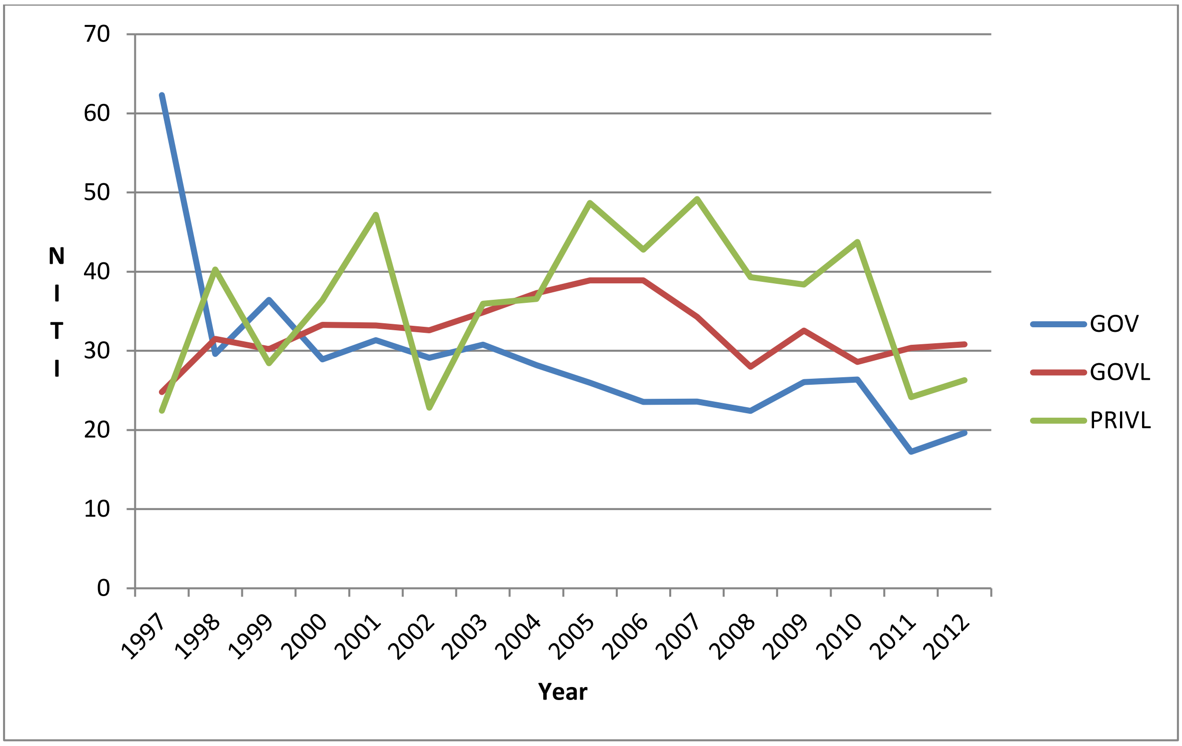


Figure 2: Non-interest Income to Total Income (NIITI): 1997-2012

This figure displays average annual non-interest income to total income (NIITI) for three sub samples from 1997 to 2012. The three sub samples included are (i) government owned banks (GOV), (ii) government owned listed banks (GOVL) and (iii) private sector listed banks (PRIVL).



Notes

[1] From a regulatory perspective, a bank is considered to be "foreign" if it is incorporated abroad and is only permitted to carry out banking operations as a branch of its overseas parent (after obtaining a licence from the Central Bank of Sri Lanka).

[2] According to the World Bank Financial Inclusion Database, more than 69% of the population above 15 years of age in Sri Lanka had bank accounts with financial institutions in 2011. This is quite substantial compared to the average of the South Asian region of 40%.

[3] The government owns the majority of outstanding shares in these banks and therefore the controlling shareholder. For example, the government has a 51% equity ownership in Housing Development Finance Corporation (HDFC).

[4] The data for foreign banks operating in Sri Lanka are not available from the BankScope database. The exclusion of these banks should not have a significant impact on our analyses because (i) these banks contribute only 12% to sector assets and (ii) their operations are mainly confined to the capital city of Colombo with the majority of them having only one branch. Additionally, many studies have reported conclusive evidence that foreign banks are much more efficient in benefiting from diversification as they belong to multinational financial conglomerates which possess expertise knowledge on diversification. The sample analysed in this study allows us to examine how domestic banks responded to opportunities provided by the liberalised economy by diversifying their income and achieving associated benefits.

[5] This is the maximum period for which the necessary data were available from the BankScope Database.

[6] Sri Lankan economy severely suffered from a nearly three-decade long separatist war which is commonly identified as the 'Ealam war' in international media. The war ended in May 2009 with the Sri Lankan forces gaining control of the last bit of territory held by the rebels, and the death of the rebel leader and other key personnel of the terrorist outfit. There had been some attempts to create a dialogue between the central government and the separatist fighters. Our sample contained some years in which war prevailed in the country, some years in which ceasefire prevailed in the country and years that represent post-war period.

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